



FROM STANDARD USER TO SYSTEM

Hunting BrokenPipe (Steam Client Service LPE)

Public PoC • No CVE Assigned • Unpatched at Time of Writing

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Environment: Isolated Windows 10 x64 Lab VM
Focus: Behavior-Based Detection Engineering

THREAT RESEARCH — DETECTION ENGINEERING

This write-up documents a publicly disclosed vulnerability for defensive purposes only. All reproduction was performed in an isolated lab environment; it does not cover weaponization.



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BrokenPipe is a publicly disclosed, unpatched Windows LPE affecting the Steam Client Service (steamservice.exe), which always runs as SYSTEM. A standard user can trigger it with no UAC prompt and no Steam account.

This post isn't about weaponizing it — it's about building behavior-based detection for this class of LPE.

1. Introduction

Most privilege escalation bugs don't get much attention on their own as an attacker already needs some way into the machine before it matters. But "local privilege escalation" (LPE) simply means: an attacker who already has low-level access (like a normal user account) finds a way to become the most powerful user on the system i.e. SYSTEM, which on Windows can do basically anything, including turning off security tools or stealing saved passwords.

That's what makes BrokenPipe worth paying attention to. Steam is installed on hundreds of millions of computers, and this bug published by researcher KillaBoi lets a normal, non-admin Windows user turn into SYSTEM just by having Steam open at its login screen. No password needed, no admin prompt, no warning popup. One moment you're a regular user, the next you have full control of the machine.

In this post, we'll not be focusing on how to exploit it, we'll focus on what matters for defenders: how do you actually catch this kind of behavior happening on a machine, using logs you probably already collected. And more importantly, how do you build a detection that would catch not just this bug, but the next LPE like it.

2. Vulnerability Overview

The Steam Client Service (steamservice.exe) is a Windows service that always runs with the highest possible privilege level i.e. SYSTEM. It exists to install and launch Steam content on behalf of whoever is logged in. To do that safely, Valve signs a special configuration file called a VDF (Valve Data Format) manifest, which is meant to ensure only trusted install scripts can be run.

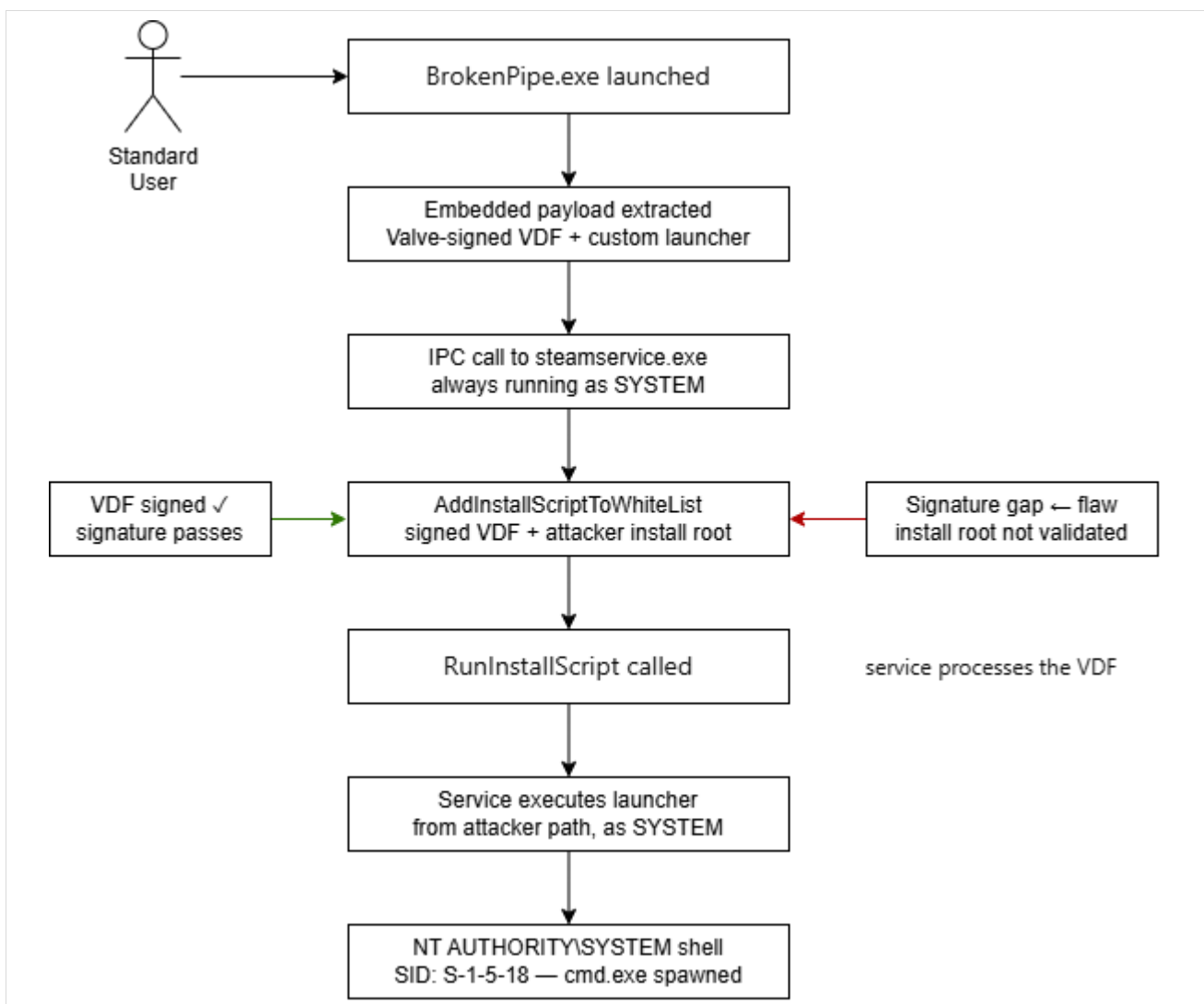
The problem is a gap in what that signature actually covers. Valve's signature verifies the contents of the VDF file, it checks that the script hasn't been tampered with. But it doesn't verify where those files will actually be executed from. That location (the installation root) can be supplied by whoever is making the request to the service, even a completely unprivileged user.



BrokenPipe exploits exactly that gap. It talks to the Steam Client Service over its own internal communication channel (IPC), points the service at a relocated path containing an attacker-controlled launcher, and because the signature check passes, the privileged SYSTEM service happily executes that launcher. No signature was forged, no bypass was performed, the flaw is simply that the signature never covered the execution path in the first place.

The result: an interactive NT AUTHORITY\SYSTEM command prompt, spawned from a standard user account with Steam sitting idle at its login screen.

3. The Attack Flow



High-level BrokenPipe attack chain; standard user to NT AUTHORITY\SYSTEM



4. Lab Setup

Everything here was built and tested in an isolated environment.

- Isolated Windows 10/11 x64 VM.
- Steam client installed, left idle at the unauthenticated login screen; no Steam account required.
- A standard (non-administrator) Windows user.
- Sysmon installed with SwiftOnSecurity's baseline config.
- Windows Security event log with process creation auditing enabled.
- Process Monitor and Process Hacker (System Informer) running alongside Sysmon for ground-truth telemetry capture.
- Splunk dev instance ingesting the Sysmon and Windows Security event logs.

5. Exploit Reproduction

Everything was reproduced in an isolated VM under a standard (non-admin) Windows user account. Before running anything: Steam is open and idle at its unauthenticated login screen, the current user confirmed as medium integrity via whoami /groups. Running the PoC requires no elevation, no UAC prompt appears, none is requested.

The result: a command prompt spawns and whoami /user returns NT AUTHORITY\SYSTEM (SID S-1-5-18). No privilege grant appears in the Windows Security event log.

```
C:\Users\Guest1\Desktop>whoami
nt authority\system

C:\Windows\system32>whoami /priv

PRIVILEGES INFORMATION
-----
Privilege Name            Description                                State
-----
SeAssignPrimaryTokenPrivilege  Replace a process level token              Disabled
SeIncreaseQuotaPrivilege      Adjust memory quotas for a process         Disabled
SeJobPrivilege                Act as part of the operating system        Enabled
SeSecurityPrivilege           Manage auditing and security log          Disabled
SeTakeOwnershipPrivilege      Take ownership of files or other objects   Disabled
SeLoadDriverPrivilege         Load and unload device drivers             Disabled
SeProfileSingleProcessPrivilege Profile single process                     Enabled
SeIncreaseBasePriorityPrivilege Increase scheduling priority               Enabled
SeCreatePermanentPrivilege    Create permanent shared objects            Enabled
SeBackupPrivilege              Back up files and directories              Disabled
SeRestorePrivilege             Restore files and directories              Disabled
SeShutdownPrivilege            Shut down the system                      Disabled
SeDebugPrivilege               Debug programs                            Enabled
SeAuditPrivilege               Generate security audits                   Enabled
SeSystemEnvironmentPrivilege  Modify firmware environment values         Disabled
SeChangeNotifyPrivilege        Bypass traverse checking                   Enabled
SeUndockPrivilege              Remove computer from docking station       Disabled
SeManageVolumePrivilege        Perform volume maintenance tasks          Disabled
SeImpersonatePrivilege          Impersonate a client after authentication   Enabled
SeCreateGlobalPrivilege        Create global objects                      Enabled
SeTrustedCredentialAccessPrivilege Access Credential Manager as a trusted caller Disabled

C:\Windows\system32>

C:\Users\Guest1\Desktop>whoami /groups

GROUP INFORMATION
-----
Group Name                Type                SID                Attributes
-----
Everyone                   Well-known group    S-1-1-0            Well-known group S-1-1-0
5-1-5-32-545 mandatory group, Enabled by default, Enabled group NT AUTHORITY\INTERACTIVE
Well-known group S-1-5-4    Mandatory group, Enabled by default, Enabled group
CONSOLE LOGON              Well-known group    S-1-2-1            Mandatory group, Enabled by default, Enabled group
NT AUTHORITY\Authenticated Users Well-known group S-1-5-11 Mandatory group, Enabled by default, Enabled group
NT AUTHORITY\This Organization Well-known group S-1-5-15 Mandatory group, Enabled by default, Enabled group
NT AUTHORITY\Local account  Well-known group S-1-5-113 Mandatory group, Enabled by default, Enabled group
LOCAL                       Well-known group    S-1-2-0            Mandatory group, Enabled by default, Enabled group
NT AUTHORITY\NTLM Authentication Well-known group S-1-5-64-10 Mandatory group, Enabled by default, Enabled group
Mandatory Label\Medium Mandatory Level Label S-1-16-8192
```

whoami /user confirming NT AUTHORITY\SYSTEM, SID S-1-5-18



Process	PPID	PID	User Name	Integrity	Path
steamservice.exe		676	NT AUTHORITY\SYSTEM	System	C:\Program Files (x86)\Common Files\Steam\steamservice.exe
launcher.exe	676	7380	NT AUTHORITY\SYSTEM	System	C:\Users\Public\BrokenPipe\runs\h1-20260917-164710-e35de8d1\host-local-pack\velocated-root\launcher.exe
cmd.exe	7380	1660	NT AUTHORITY\SYSTEM	System	C:\Windows\System32\cmd.exe
cmd.exe	1660	5484	NT AUTHORITY\SYSTEM	System	C:\Windows\System32\cmd.exe

Process Explorer showing steamservice.exe → launcher.exe → cmd.exe, all SYSTEM integrity

The detection-relevant moment isn't the exploit itself it's the integrity-level jump: steamservice.exe spawning a child process with a SYSTEM token whose image path sits outside the standard Steam install directory.

6. Telemetry Analysis

This section walks through what each stage of the exploit looks like in your telemetry, in execution order. The goal is to connect each step of the attack chain to a specific observable, so you know what to look for in Splunk, Procmon, and Process Hacker.

Phase 1 — Initial Execution

BrokenPipe.exe is executed by the standard user guest1. The binary immediately extracts its embedded resources BrokenPipe-Bootstrap.ps1 and BrokenPipe-Payload.zip to a temporary directory, then spawns powershell.exe with execution policy bypass to run the bootstrap. No elevated privileges at this point. The entire chain starts as a completely unprivileged user.

_time	EventCode	signature	user	parent_process_name	process_name	CommandLine
2026-09-17 00:29:49.424	1	Process creation	guest1	BrokenPipe.exe	powershell.exe	"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" -NoProfile -ExecutionPolicy Bypass -File "C:\Users\guest1\AppData\Local\Temp\SLTBDEC.tmp\BrokenPipe-Bootstrap.ps1" -PayloadZip "C:\Users\guest1\AppData\Local\Temp\SLTBDEC.tmp\BrokenPipePayload.zip" -ExtractRoot "C:\Users\guest1\AppData\Local\Temp\SLTBDEC.tmp\package"

BrokenPipe.exe spawns powershell.exe as guest1 with -ExecutionPolicy Bypass.

Phase 2 — Payload Staged to %TEMP% and C:\Users\Public\BrokenPipe

PowerShell unpacks BrokenPipe-Payload.zip and stages the full exploit toolkit to two locations. Temporary lab files go to AppData\Temp\SLTBDEC.tmp\. The persistent working directory C:\Users\Public\BrokenPipe\ receives the three files that matter: signed.vdf (genuine Valve-signed manifest), run.vdf (attacker-built execution script), and launcher.exe (a copy of cmd.exe). Public folder is deliberately chosen which is writable by any standard user, no permissions needed.

_time	EventCode	signature	user	process_path	process_name	file_path
2026-09-17 00:29:49.415	11	FileCreate	guest1	C:\Users\guest1\Desktop\BrokenPipe.exe	BrokenPipe.exe	C:\Users\guest1\AppData\Local\Temp\SLTBDEC.tmp\BrokenPipe-Bootstrap.ps1

BrokenPipe.exe drops BrokenPipe-Bootstrap.ps1 to AppData\Temp\



time	EventCode	signature	user	process_path	process_name	file_path
2026-09-17 00:29:50.982	11	FileCreate	guest1	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	powershell.exe	C:\Users\guest1\AppData\Local\Temp\SLT8DEC.tmp\package\steam-client-service-interactive-system-cmd-lab-10.96.30.42\Prepare-Interactive-Lab.ps1
2026-09-17 00:29:50.945	11	FileCreate	guest1	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	powershell.exe	C:\Users\guest1\AppData\Local\Temp\SLT8DEC.tmp\package\steam-client-service-interactive-system-cmd-lab-10.96.30.42\Run-Interactive-Lab.ps1
2026-09-17 00:29:51.001	11	FileCreate	guest1	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	powershell.exe	C:\Users\guest1\AppData\Local\Temp\SLT8DEC.tmp\package\steam-client-service-interactive-system-cmd-lab-10.96.30.42\assets\steam_service_interactive_demo.exe
2026-09-17 00:29:51.019	11	FileCreate	guest1	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	powershell.exe	C:\Users\guest1\AppData\Local\Temp\SLT8DEC.tmp\package\steam-client-service-interactive-system-cmd-lab-10.96.30.42\lib\steam_service_host_local_control.ps1
2026-09-17 00:29:51.042	11	FileCreate	guest1	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	powershell.exe	C:\Users\guest1\AppData\Local\Temp\SLT8DEC.tmp\package\steam-client-service-interactive-system-cmd-lab-10.96.30.42\lib\steam_service_host_local_pack.ps1
2026-09-17 00:29:51.065	11	FileCreate	guest1	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	powershell.exe	C:\Users\guest1\AppData\Local\Temp\SLT8DEC.tmp\package\steam-client-service-interactive-system-cmd-lab-10.96.30.42\lib\steam_service_ipc_frames.ps1
2026-09-17 00:29:51.110	11	FileCreate	guest1	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	powershell.exe	C:\Users\guest1\AppData\Local\Temp\SLT8DEC.tmp\package\steam-client-service-interactive-system-cmd-lab-10.96.30.42\source\steam_service_interactive_demo.vcaproj
2026-09-17 00:29:51.660	11	FileCreate	guest1	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	powershell.exe	C:\Users\Public\BrokenPipe\scratch\h1-20260916-184451-799cbc65\no-whitelist\launcher.exe
2026-09-17 00:29:51.879	11	FileCreate	guest1	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	powershell.exe	C:\Users\Public\BrokenPipe\scratch\h1-20260916-184451-799cbc65\relocated-root\launcher.exe
2026-09-17 00:29:52.091	11	FileCreate	guest1	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	powershell.exe	C:\Users\Public\BrokenPipe\runs\h1-20260916-184451-799cbc65\steam_service_host_local_control.ps1
2026-09-17 00:29:52.093	11	FileCreate	guest1	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	powershell.exe	C:\Users\Public\BrokenPipe\runs\h1-20260916-184451-799cbc65\steam_service_ipc_frames.ps1
2026-09-17 00:29:52.099	11	FileCreate	guest1	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	powershell.exe	C:\Users\Public\BrokenPipe\runs\h1-20260916-184451-799cbc65\host-local-pack\no-whitelist\launcher.exe
2026-09-17 00:29:52.101	11	FileCreate	guest1	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	powershell.exe	C:\Users\Public\BrokenPipe\runs\h1-20260916-184451-799cbc65\host-local-pack\relocated-root\launcher.exe

PowerShell stages the full exploit toolkit (launcher.exe, steam_service_ipc_frames.ps1, signed.vdf, run.vdf) to C:\Users\Public\BrokenPipe\

This PC > Local Disk (C:) > Users > Public > BrokenPipe > runs > h1-20260918-070144-a5b4a601 > host-local-pack				
Name	Date modified	Type	Size	
evidence	9/18/2026 12:46 PM	File folder		
no-whitelist	9/18/2026 12:46 PM	File folder		
relocated-root	9/18/2026 12:46 PM	File folder		
signature-negative	9/18/2026 12:46 PM	File folder		
signed-input	9/18/2026 12:46 PM	File folder		
manifest	9/18/2026 12:46 PM	JSON Source File	14 KB	
marker	9/18/2026 12:46 PM	JSON Source File	1 KB	

All files dropped to C:\Users\Public\BrokenPipe

Phase 3 — IPC Framework Loaded and Whitelist IPC Call

PowerShell loads steam_service_ipc_frames.ps1 (the library that constructs and sends IPC messages to the Steam service). Script block logging (EID 4104) captures its full content before any IPC call fires. This reveals the method opcodes that identify each IPC call: 0x3f for the whitelist request and 0x41 for the run request.

```
$script:IpOuterCommand = [byte]0x01
$script:IpInterfaceSelector = [byte]0x01
$script:IpSteamUser = [uint32]0
$script:IpWhitelistMethod = [byte]0x3f
$script:IpRunMethod = [byte]0x41
$script:IpQueryMethod = [byte]0x42
$script:IpWhitelistFencepost = [uint32]0xbe
$script:IpRunFencepost = [uint32]0xc0
$script:IpQueryFencepost = [uint32]0xc1
```

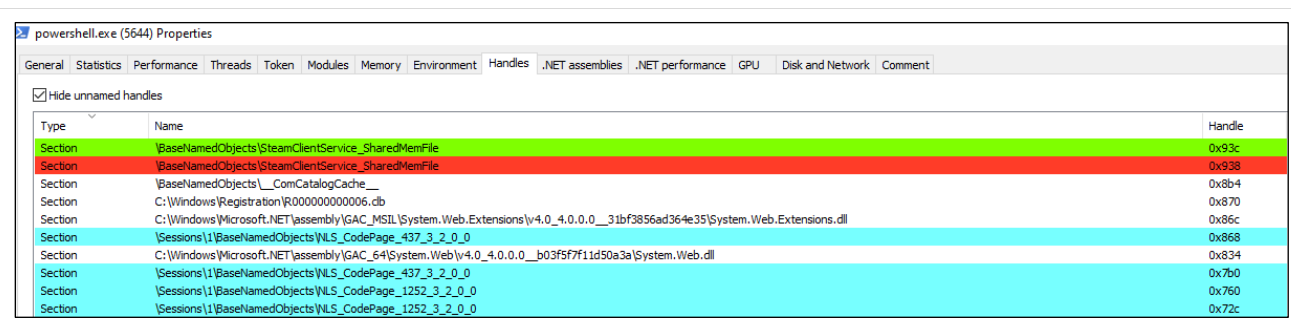
IPC opcodes captured in the script block log. 0x3f = whitelist call, 0x41 = run call.



The New-IpcWhitelistPayload function constructs the whitelist request with two fields: the path to signed.vdf (a genuine Valve signature that passes validation) and C:\Users\Public\BrokenPipe\ as the installation root (an attacker-controlled path not covered by the signature). Although the Steam service validates the signature successfully, it blindly trusts the installation root, allowing launcher.exe from the attacker-controlled path to be whitelisted. This is the vulnerability: one field is validated, while the other is trusted without proper verification.

Phase 4 — IPC Channel Confirmed Open

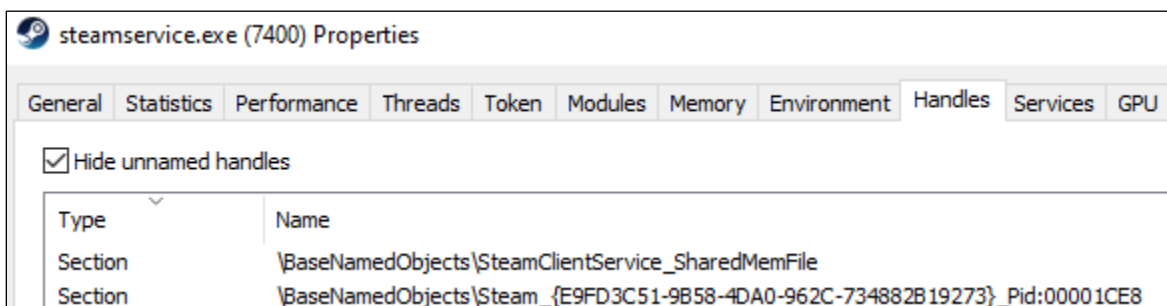
Process Hacker reveals powershell.exe holds an open handle to SteamClientService_SharedMemFile. This is the shared memory object used as the IPC channel to steamservice.exe. It requires no admin privilege to open. Any standard user process can reach it.



Type	Name	Handle
Section	\BaseNamedObjects\SteamClientService_SharedMemFile	0x93c
Section	\BaseNamedObjects\SteamClientService_SharedMemFile	0x938
Section	\BaseNamedObjects_ComCatalogCache_	0x8b4
Section	C:\Windows\Registration\R0000000000006.cb	0x870
Section	C:\Windows\Microsoft.NET\assembly\GAC_MSIL\System.Web.Extensions\v4.0.0.0__31bf3856ad364e35\System.Web.Extensions.dll	0x86c
Section	\Sessions\1\BaseNamedObjects\NLS_CodePage_437_3_2_0_0	0x868
Section	C:\Windows\Microsoft.NET\assembly\GAC_64\System.Web\v4.0.0.0__b03f5f7f11d50a3a\System.Web.dll	0x834
Section	\Sessions\1\BaseNamedObjects\NLS_CodePage_437_3_2_0_0	0x7b0
Section	\Sessions\1\BaseNamedObjects\NLS_CodePage_1252_3_2_0_0	0x760
Section	\Sessions\1\BaseNamedObjects\NLS_CodePage_1252_3_2_0_0	0x72c

powershell.exe handles tab confirming the shared memory IPC object

On the service side, the same shared memory object is held by steamservice.exe (PID 7400) simultaneously. Both ends of the channel are visible at the same time, one on the attacker's process, one on the SYSTEM service confirming the IPC handshake completed successfully.



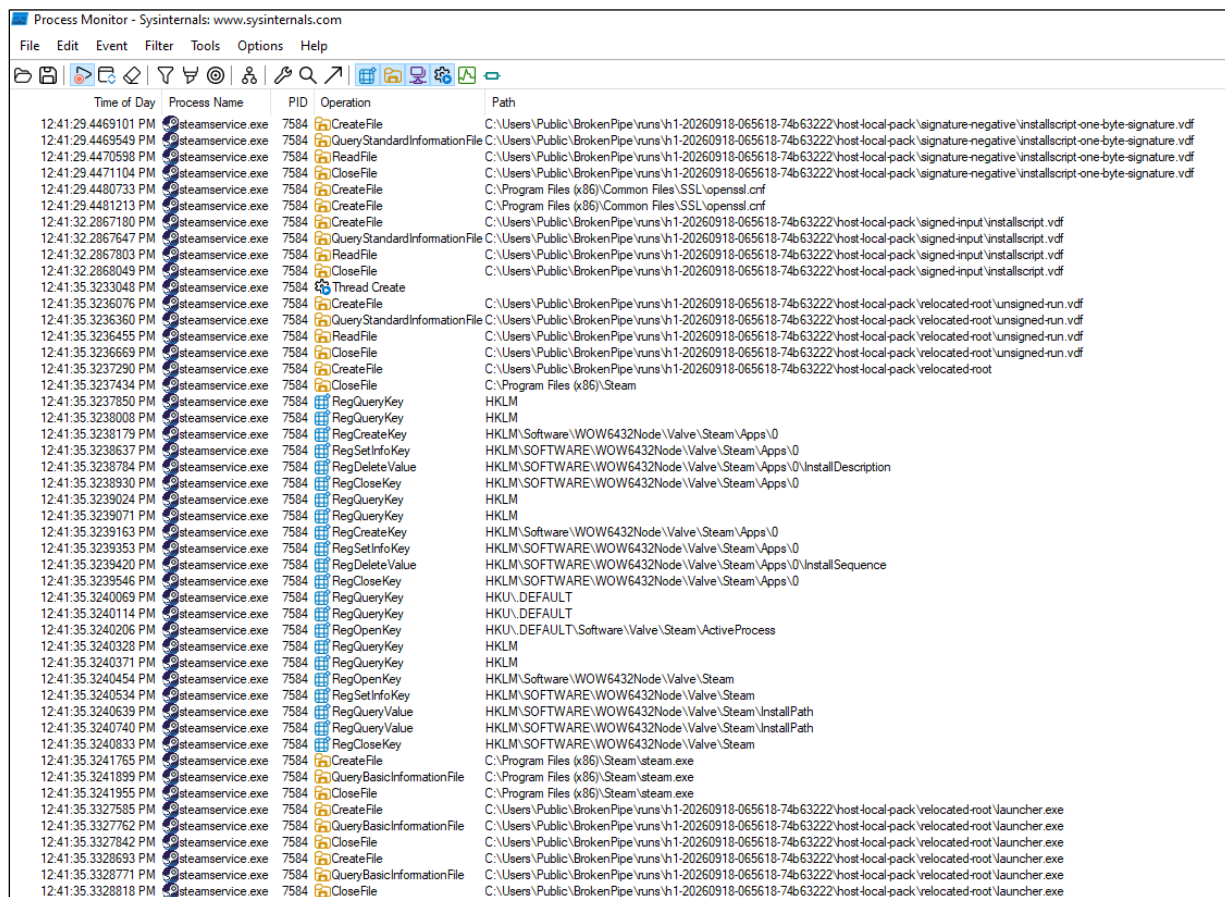
Type	Name
Section	\BaseNamedObjects\SteamClientService_SharedMemFile
Section	\BaseNamedObjects\Steam_{E9FD3C51-9B58-4DA0-962C-734882B19273}_Pid:00001CE8

steamservice.exe handles tab confirming both ends of the shared memory object



Phase 5 — Service Accepts and Executes

Procmon trace confirms the moment the channel is established: steamservice.exe immediately begins file and registry operations against the BrokenPipe working directory, showing the service has received and is processing the incoming IPC request from the standard user process. steamservice.exe validates the VDF contents but the signature passes. The WorkRoot path (C:\Users\Public\BrokenPipe\) is accepted without any validation. This is the entire flaw one field validated, the other trusted completely.

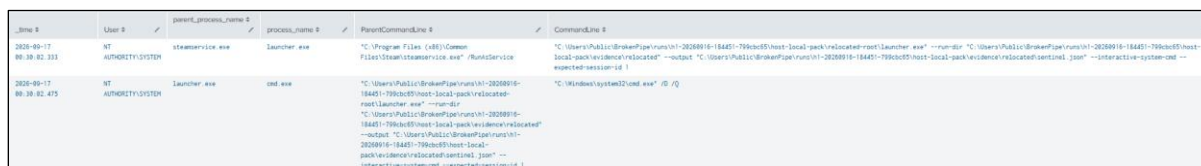


Time of Day	Process Name	PID	Operation	Path
12:41:29.4469101 PM	steamservice.exe	7584	CreateFile	C:\Users\Public\BrokenPipe\vsns\h1-20260918-065618-74b63222\host-local-pack\signature-negative\installscript-one-byte-signature.vdf
12:41:29.4469549 PM	steamservice.exe	7584	QueryStandardInformationFile	C:\Users\Public\BrokenPipe\vsns\h1-20260918-065618-74b63222\host-local-pack\signature-negative\installscript-one-byte-signature.vdf
12:41:29.4470598 PM	steamservice.exe	7584	ReadFile	C:\Users\Public\BrokenPipe\vsns\h1-20260918-065618-74b63222\host-local-pack\signature-negative\installscript-one-byte-signature.vdf
12:41:29.4471104 PM	steamservice.exe	7584	CloseFile	C:\Users\Public\BrokenPipe\vsns\h1-20260918-065618-74b63222\host-local-pack\signature-negative\installscript-one-byte-signature.vdf
12:41:29.4480733 PM	steamservice.exe	7584	CreateFile	C:\Program Files (x86)\Common Files\SSL\openssl.cnf
12:41:29.4481213 PM	steamservice.exe	7584	CreateFile	C:\Program Files (x86)\Common Files\SSL\openssl.cnf
12:41:32.2867180 PM	steamservice.exe	7584	CreateFile	C:\Users\Public\BrokenPipe\vsns\h1-20260918-065618-74b63222\host-local-pack\signed-input\installscript.vdf
12:41:32.2867647 PM	steamservice.exe	7584	QueryStandardInformationFile	C:\Users\Public\BrokenPipe\vsns\h1-20260918-065618-74b63222\host-local-pack\signed-input\installscript.vdf
12:41:32.2867803 PM	steamservice.exe	7584	ReadFile	C:\Users\Public\BrokenPipe\vsns\h1-20260918-065618-74b63222\host-local-pack\signed-input\installscript.vdf
12:41:32.2868049 PM	steamservice.exe	7584	CloseFile	C:\Users\Public\BrokenPipe\vsns\h1-20260918-065618-74b63222\host-local-pack\signed-input\installscript.vdf
12:41:35.3233048 PM	steamservice.exe	7584	Thread Create	
12:41:35.3236076 PM	steamservice.exe	7584	CreateFile	C:\Users\Public\BrokenPipe\vsns\h1-20260918-065618-74b63222\host-local-pack\velocated-root\unsigned-run.vdf
12:41:35.3236360 PM	steamservice.exe	7584	QueryStandardInformationFile	C:\Users\Public\BrokenPipe\vsns\h1-20260918-065618-74b63222\host-local-pack\velocated-root\unsigned-run.vdf
12:41:35.3236455 PM	steamservice.exe	7584	ReadFile	C:\Users\Public\BrokenPipe\vsns\h1-20260918-065618-74b63222\host-local-pack\velocated-root\unsigned-run.vdf
12:41:35.3236669 PM	steamservice.exe	7584	CloseFile	C:\Users\Public\BrokenPipe\vsns\h1-20260918-065618-74b63222\host-local-pack\velocated-root\unsigned-run.vdf
12:41:35.3237290 PM	steamservice.exe	7584	CreateFile	C:\Users\Public\BrokenPipe\vsns\h1-20260918-065618-74b63222\host-local-pack\velocated-root\unsigned-run.vdf
12:41:35.3237434 PM	steamservice.exe	7584	CloseFile	C:\Program Files (x86)\Steam
12:41:35.3237850 PM	steamservice.exe	7584	RegQueryKey	HKLM
12:41:35.3238008 PM	steamservice.exe	7584	RegQueryKey	HKLM
12:41:35.3238179 PM	steamservice.exe	7584	RegCreateKey	HKLM\Software\WOW6432Node\Valve\Steam\Apps\0
12:41:35.3238637 PM	steamservice.exe	7584	RegSetInfoKey	HKLM\SOFTWARE\WOW6432Node\Valve\Steam\Apps\0
12:41:35.3238784 PM	steamservice.exe	7584	RegDeleteValue	HKLM\SOFTWARE\WOW6432Node\Valve\Steam\Apps\0\InstallDescription
12:41:35.3238930 PM	steamservice.exe	7584	RegCloseKey	HKLM\SOFTWARE\WOW6432Node\Valve\Steam\Apps\0
12:41:35.3239024 PM	steamservice.exe	7584	RegQueryKey	HKLM
12:41:35.3239071 PM	steamservice.exe	7584	RegQueryKey	HKLM
12:41:35.3239163 PM	steamservice.exe	7584	RegCreateKey	HKLM\Software\WOW6432Node\Valve\Steam\Apps\0
12:41:35.3239353 PM	steamservice.exe	7584	RegSetInfoKey	HKLM\SOFTWARE\WOW6432Node\Valve\Steam\Apps\0
12:41:35.3239420 PM	steamservice.exe	7584	RegDeleteValue	HKLM\SOFTWARE\WOW6432Node\Valve\Steam\Apps\0\InstallSequence
12:41:35.3239546 PM	steamservice.exe	7584	RegCloseKey	HKLM\SOFTWARE\WOW6432Node\Valve\Steam\Apps\0
12:41:35.3240069 PM	steamservice.exe	7584	RegQueryKey	HKU\DEFAULT
12:41:35.3240114 PM	steamservice.exe	7584	RegQueryKey	HKU\DEFAULT
12:41:35.3240206 PM	steamservice.exe	7584	RegOpenKey	HKU\DEFAULT\Software\Valve\Steam\ActiveProcess
12:41:35.3240328 PM	steamservice.exe	7584	RegQueryKey	HKLM
12:41:35.3240371 PM	steamservice.exe	7584	RegQueryKey	HKLM
12:41:35.3240454 PM	steamservice.exe	7584	RegOpenKey	HKLM\Software\WOW6432Node\Valve\Steam
12:41:35.3240534 PM	steamservice.exe	7584	RegSetInfoKey	HKLM\SOFTWARE\WOW6432Node\Valve\Steam
12:41:35.3240639 PM	steamservice.exe	7584	RegQueryValue	HKLM\SOFTWARE\WOW6432Node\Valve\Steam\InstallPath
12:41:35.3240740 PM	steamservice.exe	7584	RegQueryValue	HKLM\SOFTWARE\WOW6432Node\Valve\Steam\InstallPath
12:41:35.3240833 PM	steamservice.exe	7584	RegCloseKey	HKLM\SOFTWARE\WOW6432Node\Valve\Steam
12:41:35.3241765 PM	steamservice.exe	7584	CreateFile	C:\Program Files (x86)\Steam\steam.exe
12:41:35.3241899 PM	steamservice.exe	7584	QueryBasicInformationFile	C:\Program Files (x86)\Steam\steam.exe
12:41:35.3241955 PM	steamservice.exe	7584	CloseFile	C:\Program Files (x86)\Steam\steam.exe
12:41:35.3247585 PM	steamservice.exe	7584	CreateFile	C:\Users\Public\BrokenPipe\vsns\h1-20260918-065618-74b63222\host-local-pack\velocated-root\launcher.exe
12:41:35.3247762 PM	steamservice.exe	7584	QueryBasicInformationFile	C:\Users\Public\BrokenPipe\vsns\h1-20260918-065618-74b63222\host-local-pack\velocated-root\launcher.exe
12:41:35.3247842 PM	steamservice.exe	7584	CloseFile	C:\Users\Public\BrokenPipe\vsns\h1-20260918-065618-74b63222\host-local-pack\velocated-root\launcher.exe
12:41:35.3248933 PM	steamservice.exe	7584	CreateFile	C:\Users\Public\BrokenPipe\vsns\h1-20260918-065618-74b63222\host-local-pack\velocated-root\launcher.exe
12:41:35.3248771 PM	steamservice.exe	7584	QueryBasicInformationFile	C:\Users\Public\BrokenPipe\vsns\h1-20260918-065618-74b63222\host-local-pack\velocated-root\launcher.exe
12:41:35.3248818 PM	steamservice.exe	7584	CloseFile	C:\Users\Public\BrokenPipe\vsns\h1-20260918-065618-74b63222\host-local-pack\velocated-root\launcher.exe

steamservice.exe responding to the IPC request; CreateFile, ReadFile, and registry operations begin immediately after the shared memory channel is established.

Phase 6 — SYSTEM Shell Confirmed

Sysmon EID 1 fires, steamservice.exe spawning cmd.exe with User: NT AUTHORITY\SYSTEM and IntegrityLevel: System. Standard user guest1 has a SYSTEM shell. No admin credentials. No UAC prompt.



Time	User	parent_process_name	process_name	ParentCommandLine	CommandLine
2025-09-17 09:38:02.333	NT AUTHORITY\SYSTEM	steamservice.exe	launcher.exe	"C:\Program Files (x86)\Common Files\Steam\steamservice.exe" /RunService	"C:\Users\Public\BrokenPipe\vsns\h1-20260918-065618-74b63222\host-local-pack\velocated-root\launcher.exe" --number "C:\Users\Public\BrokenPipe\vsns\h1-20260918-065618-74b63222\host-local-pack\velocated-root\launcher.exe" --output "C:\Users\Public\BrokenPipe\vsns\h1-20260918-065618-74b63222\host-local-pack\velocated-root\launcher.exe" --expectedversionid 1
2025-09-17 09:38:02.475	NT AUTHORITY\SYSTEM	launcher.exe	cmd.exe	"C:\Users\Public\BrokenPipe\vsns\h1-20260918-065618-74b63222\host-local-pack\velocated-root\launcher.exe" --number "C:\Users\Public\BrokenPipe\vsns\h1-20260918-065618-74b63222\host-local-pack\velocated-root\launcher.exe" --output "C:\Users\Public\BrokenPipe\vsns\h1-20260918-065618-74b63222\host-local-pack\velocated-root\launcher.exe" --expectedversionid 1	"C:\Windows\system32\cmd.exe" /Q

steamservice.exe spawning launcher.exe from C:\Users\Public\BrokenPipe\



Time #	ParentPID #	PID #	User #	parent_process_name #	process_name #	ParentCommandLine #	CommandLine #
2025-09-17 00:30:02.333	4836	6680	NT AUTHORITY\SYSTEM	steamservice.exe	launcher.exe	"C:\Program Files (x86)\Common Files\Steam\steamservice.exe" /RunksService	"C:\Users\Public\BrokenPipe\runt\h1-20260916-184451-799cb65\host-local-pack\relocated-root\launcher.exe" --run-dir "C:\Users\Public\BrokenPipe\runt\h1-20260916-184451-799cb65\host-local-pack\evdence\relocated" --output "C:\Users\Public\BrokenPipe\runt\h1-20260916-184451-799cb65\host-local-pack\evdence\relocated\sentinel.json" --interactive-system-cmd --expected-session-id 1
2025-09-17 00:30:02.475	6680	6616	NT AUTHORITY\SYSTEM	launcher.exe	cmd.exe	"C:\Users\Public\BrokenPipe\runt\h1-20260916-184451-799cb65\host-local-pack\relocated-root\launcher.exe" --run-dir "C:\Users\Public\BrokenPipe\runt\h1-20260916-184451-799cb65\host-local-pack\evdence\relocated" --output "C:\Users\Public\BrokenPipe\runt\h1-20260916-184451-799cb65\host-local-pack\evdence\relocated\sentinel.json" --interactive-system-cmd --expected-session-id 1	"C:\Windows\system32\cmd.exe" /D /Q
2025-09-17 00:30:11.707	6616	976	NT AUTHORITY\SYSTEM	cmd.exe	whoami.exe	"C:\Windows\system32\cmd.exe" /D /Q	whoami
2025-09-17 00:30:15.892	6616	6396	NT AUTHORITY\SYSTEM	cmd.exe	whoami.exe	"C:\Windows\system32\cmd.exe" /D /Q	whoami /priv
2025-09-17 00:30:20.193	6616	2216	NT AUTHORITY\SYSTEM	cmd.exe	whoami.exe	"C:\Windows\system32\cmd.exe" /D /Q	whoami /groups

Final SYSTEM shell; steamservice.exe → launcher.exe → cmd.exe chain confirmed

7. Detection & Hunting Queries

7.1 Hunting Queries in Splunk

Hunt 1: SYSTEM Child Process Execution from Non-OS Service Paths

```
index=main EventCode=1
IntegrityLevel="System"
ParentImage IN ("*steamservice.exe","*updater.exe","*launcher.exe","*agent.exe","*helper.exe")
NOT Image IN (
  "C:\\Program Files\\*",
  "C:\\Program Files (x86)\\*",
  "C:\\Windows\\*"
)
| table _time, ComputerName, User, ParentImage, Image, CommandLine, IntegrityLevel
| sort - _time
```

Hunt 2: PowerShell Writing Executables to User-Writable Public Paths

```
index=main EventCode=11
Image="*\\powershell.exe"
NOT TargetFilename="*\\_PSScriptPolicyTest_*.ps1"
(TargetFilename="C:\\Users\\Public\\*" OR TargetFilename="*\\AppData\\*" OR
TargetFilename="*\\Temp\\*")
(TargetFilename="*.exe" OR TargetFilename="*.vdf" OR TargetFilename="*.ps1")
| table _time, ComputerName, User, Image, TargetFilename
| sort - _time
```



7.2 Sigma Rules

Rule 1: SYSTEM Service Spawns Child Process Outside Install Directory

```
title: BrokenPipe - Steam LPE SYSTEM Child Outside Install Directory
id: 895d3358-e983-44bf-a69f-0577449c177c
status: experimental
description: >
  Detects steamservice.exe spawning a child process with SYSTEM integrity
  from an image path outside the standard Steam install directory.
references:
  - https://github.com/KillaBoi/BrokenPipe
author: Saksham Bhattarai
date: 2026-09-19
tags:
  - attack.privilege_escalation
  - attack.t1548
  - attack.t1543.003
  - attack.t1574
logsource:
  category: process_creation
  product: windows
detection:
  selection_parent:
    ParentImage|endswith: '\steamservice.exe'
  selection_integrity:
    IntegrityLevel: 'System'
  filter_main_legit_paths:
    Image|startswith:
      - 'C:\Program Files (x86)\Steam\'
      - 'C:\Program Files\Steam\'
      - 'C:\Windows\System32\'
  condition: selection_parent and selection_integrity and not filter_main_legit_paths
falsepositives:
  - Legitimate Steam updates with non-default install paths
  - Custom Steam library locations outside Program Files
level: high
```



Rule 2: PowerShell Dropping Executables to Public/User Paths

```
title: PowerShell Dropping Executable to User-Writable Directory
id: c2f7fc01-c538-4473-abc8-457b02f8e8ca
status: experimental
description: >
    Detects PowerShell writing executable or script files to user-writable
    paths such as C:\Users\Public or TEMP directories. May indicate
    payload staging prior to execution as seen in BrokenPipe-style LPE chains.
author: Saksham Bhattarai
date: 2026-09-19
tags:
    - attack.execution
    - attack.t1059.001
    - attack.t1105
logsource:
    category: file_event
    product: windows
detection:
    selection_process:
        Image|endswith:
            - '\powershell.exe'
            - '\pwsh.exe'
    selection_path:
        TargetFilename|contains:
            - '\Users\Public\'
            - '\AppData\Local\Temp\'
    selection_ext:
        TargetFilename|endswith:
            - '.exe'
            - '.ps1'
    filter_main_policy_test:
        TargetFilename|re: '\\__PSScriptPolicyTest_[^\\]+\.ps1$'
    condition: selection_process and selection_path and selection_ext and not filter_main_policy_test
falsepositives:
    - Legitimate software deployment scripts
    - Admin scripts that stage tools to Public folder
    - Package managers extracting to Temp
level: medium
```



8. MITRE ATT&CK Mapping

Tactic	Technique ID	Name	Description
Privilege Escalation	T1543.003	Create or Modify System Process: Windows Service	Abuses steamservice.exe (always-running SYSTEM service) to execute attacker-controlled binary under its own token
Privilege Escalation	T1574	Hijack Execution Flow	Supplies attacker-controlled install root to redirect service execution to an untrusted user-writable path
Execution	T1059.001	PowerShell	Full exploit chain runs as PowerShell scripts with -ExecutionPolicy Bypass
Execution	T1105	Ingress Tool Transfer	Payload ZIP staged to C:\Users\Public\BrokenPipe\ dropped before IPC begins
Defense Evasion	T1548	Abuse Elevation Control Mechanism	Bypasses UAC entirely (no prompt, no consent, no admin credential at any point)

9. Conclusion

BrokenPipe doesn't look like much on paper, no memory corruption, no kernel exploit, just one path that nobody validated. That's exactly what makes it worth paying attention to. Simple, reliable, and works on every machine with Steam installed. The real takeaway isn't about Steam specifically. Privileged services that trust caller-controlled input inside a signed container is a pattern that shows up across auto-updaters, game launchers, and enterprise agents from dozens of vendors. Valve happened to be the one with a public PoC. They won't be the last.

10. References

- [KillaBoi/BrokenPipe](#)
- [Video walkthrough](#)
- [MITRE ATT&CK — Privilege Escalation](#)